



Robust roll angular error measurement system for precision machines

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Abstract: This paper presents a robust and low-cost roll angle measurement system (RAMS) on the basis of two parallel beams in association with two position detectors. The commonly occurring influences of beam drift, beam diameter, and intensity variations, and non-parallelism of dual-beam on the roll angular error measurement of precision linear stages are thoroughly considered and reduced. The effectiveness of the designed system and proposed methods were demonstrated by a series of experiments. It has been verified that the designed system's measurement accuracy is within ± 1.2 arcsec over a measurement range of 1 m. The designed system is easy to construct both in the laboratory environment and factory field.

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1. Introduction

There is an increasing demand for higher accuracy and resolution in precision machines, such as numerically controlled (NC) machine tools [1,2], coordinate measuring machines (CMMs) [3] and wafer stages [4]. However, due to the manufacturing error and assemble error, there are six geometric errors existed in each linear stage, namely one positioning error, two straightness errors (vertical and horizontal straightness errors) and three angular errors (pitch, yaw and roll). Measurement of these geometric errors is an essential task for accuracy calibration and error compensation of precision machines.

Optical measurement methods, with the advantages of non-contact, high-sensitivity and fast speed, play an important role in the measurement of geometric errors of linear stages. Among those six geometric errors, small angular error in roll is conventionally detected by the techniques based on the principle of optoelectronic level [5], interference [6,7], variation of laser polarization state [8,9], and relative position shift of two parallel laser beams [10–15]. The level requires settling time before steady-state output so that real-time measurement is not possible. The interference method, although has high resolution, is complicated in system construction and has not popularly used in industry yet. The variation of polarization state is not sensitive enough and the resolution is low in detecting the intensity change of the photodetector. The simplest and on-line applicable one is to detect the relative position shift of two parallel laser beams using two position-sensitive detectors (PSDs). It is known that, no matter using a large He-Ne laser tube or a small laser diode, all laser sources have an inevitable common problem on the angular drift of the laser beam [2,16]. In addition, to generate two ideally parallel laser beams in space is a difficult task in practice. Moreover, the spot size and intensity of the laser beam would vary with its propagation distance, yielding the variation of the detector sensitivity [17,18]. These problems from the laser source that would influence the roll angle measurement have rarely been addressed and resolved before. Therefore, an effective method is necessary that considers and solves the above-mentioned influences of the laser source on the roll angle measurement.

In this report, a robust, compact and low-cost roll angle measurement system (RAMS), which can compensate for the errors induced by the beam drift, beam diameter and intensity variations and non-parallelism of dual-beam, is designed. The principle of roll angle measurement and methods for error compensation are presented. A prototype system was constructed. Experiments verified the applicability and robustness of the system.

2. Methodology

A roll angle measurement system (RAMS), as shown in Fig. 1, is designed for simultaneous measurement of the two straightness errors (δ_x and δ_y) and one roll angular error (θ_r) of a precision linear stage, being a 3 degree-of-freedom (3DOF) measuring system. The RAMS is composed of an optical sensor head and a detecting unit. The optical sensor head is fixed to the machine base and kept stationary after careful beam axis alignment to the linear stage. The detecting unit is mounted on the moving table of the linear stage.

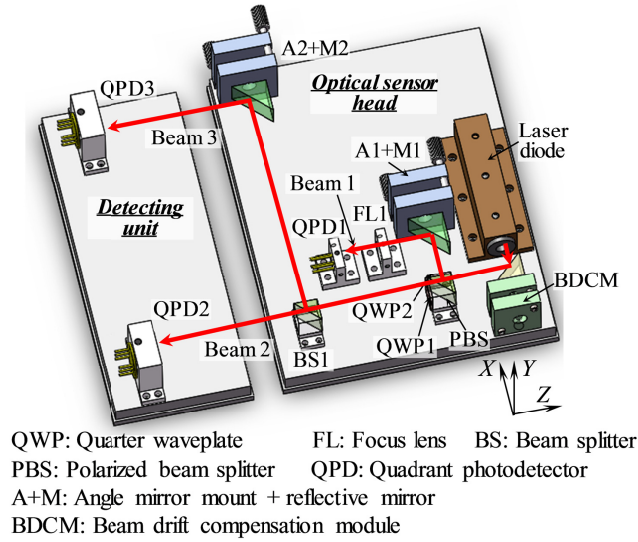


Fig. 1. Schematic of the RAMS for precision machines.

A collimated laser beam emitted from a low-cost laser diode is projected onto the mirror of a beam drift compensation module (BDCM), which is used for compensating angular errors induced by laser beam drift. The reflected beam is divided into two beams (reflected S-polarized beam and transmitted P-polarized beam) by a polarizing beam splitter (PBS). The reflected S-polarized beam (beam 1) passes through a quarter waveplate (QWP2) and enters into an autocollimator set, consisting of a focus lens (FL1) and a quadrant-photodetector (QPD1), for detecting the beam drifted angles in the pitch and yaw directions. The transmitted P-polarized beam passes through QWP1 and is split to two beams by a beam splitter (BS1). The transmitted beam (beam 2) is then projected onto QPD2 for measuring the straightness errors (δ_x and δ_y). The reflected beam (beam 3), bent by a reflective mirror (M2), is received by QPD3 for roll angular error measurement (θ_r). It should be noted that the unit, consisting of PBS, QWP1 and QWP2, will change the polarization state of the laser beam, so that the reflected beams would be stopped to go back into the laser diode.

2.1 Measurement principle of roll angular error

The measurement principle of straightness errors is to detect two lateral motions of QPD2 (Δx_2 and Δy_2), which is mounted on the moving table, with respect to the collimated laser beam, as shown in Fig. 2(a). Thus, the straightness errors (δ_x and δ_y) can be evaluated by

$$\delta_x = \Delta x_2 = k_{x2} \cdot \frac{(I_{A2} + I_{D2}) - (I_{B2} + I_{C2})}{I_{A2} + I_{B2} + I_{C2} + I_{D2}} = k_{x2} \cdot \Delta I_{x2} \quad (1)$$

$$\delta_y = \Delta y_2 = k_{y2} \cdot \frac{(I_{A2} + I_{B2}) - (I_{C2} + I_{D2})}{I_{A2} + I_{B2} + I_{C2} + I_{D2}} = k_{y2} \cdot \Delta I_{y2} \quad (2)$$

where, k_{x2} and k_{y2} are the X - and Y -directional sensitivities of QPD2, respectively. I_{A2} , I_{B2} , I_{C2} and I_{D2} indicate the photoelectric current outputs of QPD2's four quadrants, respectively. ΔI_{x2} and ΔI_{y2} are output photocurrent corresponding to Δx_2 and Δy_2 , respectively. Without further description, k_{mj} and ΔI_{mi} ($m = x, y$ and $j = 1, 2, 3$) in following text represent the sensitivity of QPD j in the X - and Y -directions, and the output photocurrent corresponding to Δm_i , respectively.

In order to measure the roll angular error, beam 2 and beam 3, shown in Fig. 1, should be carefully adjusted to be parallel to each other, especially in the Y -direction. The measurement principle of roll angular error is to detect the differential output of QPD2 (Δy_2) and QPD 3 (Δy_3), as show in Fig. 2(b). If the moving table has no roll angular error, the variation of two QPDs' outputs should be the same. Therefore, the roll angular error can be calculated by the following equation,

$$\theta_{roll} = \frac{\Delta y_2 - \Delta y_3}{L} = \frac{1}{L} (k_{y2} \cdot \Delta I_{y2} - k_{y3} \cdot \Delta I_{y3}) \quad (3)$$

where, L is the distance between beam 2 and beam 3.

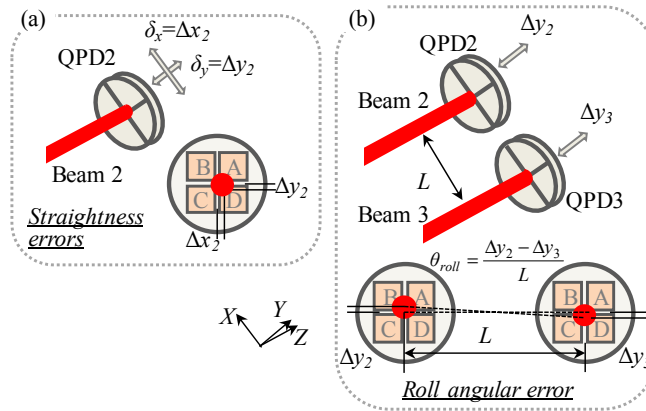


Fig. 2. Measurement principles of (a) straightness errors and (b) roll angular error.

2.2 Correction of detector sensitivity

According to Eq. (1) to Eq. (3), the sensitivity of QPDs calibrated by a displacement sensor are essential to measure the straightness errors and the roll angular error. In most cases, a constant detector sensitivity is used over the entire measurement range [2,10–15]. However, it is known that the spot size and intensity of a laser beam will vary with its propagation distance [17], resulting in the variation of detector sensitivity. It has been experimentally verified that the sensitivity of QPD is in linear relationship with the measurement distance [18]. Therefore, in order to reduce the effects of the spot size and intensity, calibration tests of a QPD can be carried out at two distant positions, which are selected as the starting position (d_o) and the final position (d_f) of the travel in this study. Let the calibrated sensitivities at d_o and d_f be defined as k_{mi_o} and k_{mi_f} ($m = x, y$ and $i = 2, 3$), respectively. Thus, the sensitivity at any other position d_z between d_o and d_f can be obtained by linear interpolation as

$$k_{mi_z} = \frac{k_{mi_f} - k_{mi_o}}{d_f - d_o} \cdot d_z - \frac{k_{mi_f} - k_{mi_o}}{d_f - d_o} \cdot d_o + k_{mi_o} \quad m = x, y \quad i = 2, 3 \quad (4)$$

The error induced by detector sensitivity variation in measurements of the straightness errors and the roll angular error can thus be theoretically reduced by Eq. (4).

2.3 Compensation of non-parallelism of dual-beam

As mentioned above, it is difficult to generate two ideally parallel laser beams. Therefore, a compensation method is proposed for the error caused by non-parallelism of dual-beam, as shown in Fig. 3. A high precision commercial electronic level is taken as a reference in this method. Particular effort is put into aligning beam 2 with the moving axis of the linear stage, so that the laser spot of beam 2 can be always close to the center of QPD2 during the motion of the detecting unit. Similarly, the laser spot of beam 3 also has to be adjusted to close to the center of QPD3 throughout the travel of the moving table. The detecting unit is firstly moved to d_o , as shown in Fig. 3(a), at which roll angular error of the linear stage is measured to be θ_{o_R} and θ_{o_L} by using the RAMS and the electronic level, respectively. Then, the detecting unit is positioned to d_f and the roll angular error is measured to be θ_{f_R} and θ_{f_L} , as shown in Fig. 3(b). Theoretically, the roll angular error measured by the RAMS and the electronic level should be the same. However, due to the non-parallelism of dual-beam, there is an error θ_p existed in the outputs of the RAMS, which can be expressed by

$$\theta_{o_R} = \frac{(\Delta y_{2_o} - \Delta y_{3_o}) + \theta_p \cdot d_o}{L} = \theta_{o_L} + \frac{\theta_p \cdot d_o}{L} \quad (5)$$

$$\theta_{f_R} = \frac{(\Delta y_{2_f} - \Delta y_{3_f}) + \theta_p \cdot d_f}{L} = \theta_{f_L} + \frac{\theta_p \cdot d_f}{L} \quad (6)$$

The non-parallel error θ_p can be expressed by

$$\theta_p = \frac{(\theta_{f_R} - \theta_{f_L}) - (\theta_{o_R} - \theta_{o_L})}{d_f - d_o} \cdot L \quad (7)$$

Thus, the actual roll angular error θ_z at any other position d_z can be compensated to be

$$\theta_z(z) = \theta_{z_R} - \frac{\theta_p \cdot d_z}{L} \quad (8)$$

where, θ_{z_R} is the roll angular error measured by the RAMS and θ_p is a constant once it is obtained by Eq. (7).

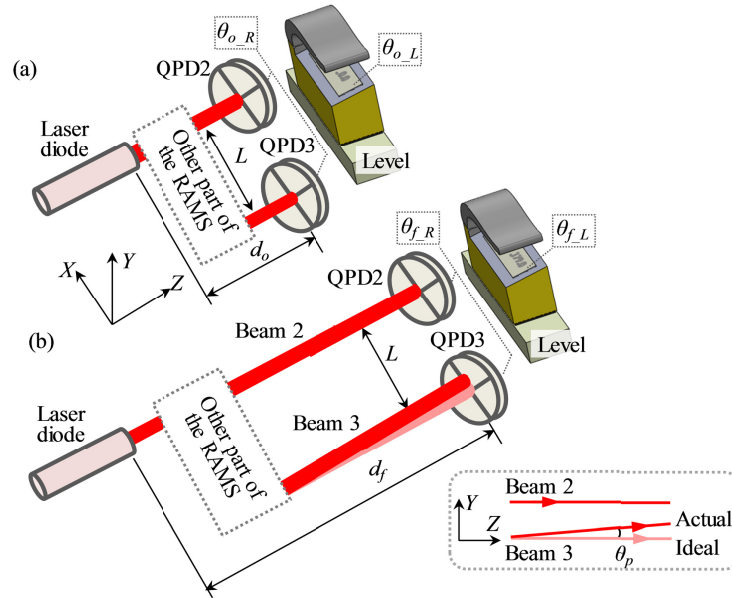


Fig. 3. Compensation principle for error induced by non-parallelism of dual-beam at (a) d_o and (b) d_f .

2.4 Compensation of laser beam drift

Laser beam drift has been recognized as one of critical error sources in the measurements of straightness and angular errors [2]. A compensation method using a beam drift compensation module (BDCM) has been developed by the authors' group [2], as shown in Fig. 4. The BDCM consists of an angle mirror mount, a right-angle prism mirror, two mini-sized PZT actuators and two pins. The pin and mini-sized PZT actuator are sequentially inserted into a large hole, which is drilled in the front of the angle mirror mount's threaded shaft. A small hole is then drilled in the end of the threaded shaft, so that the wires of the PZT actuators can be connected to the PZT driver. The right-angle prism mirror is glued on the angle mirror mount to provide adjustment of laser beam in the pitch and yaw directions. This integrated BDCM can provide not only large manual angle adjustment to assist beam alignment procedure but also fine PZT tuning to real-time compensate the beam drift in association with the following autocollimator set. The autocollimator set, consisting of QPD1 and FL1 with a focal length of f , is constructed to detect drifted angles (θ_{pitch} and θ_{yaw}) based on the principle of 2D autocollimator, as shown in the upper right corner of Fig. 4. Assuming that θ_{pitch} and θ_{yaw} induced linear displacements of the laser spot on QPD1 are denoted by Δx_1 and Δy_1 , which can be obtained from the photoelectric current outputs of QPD1, θ_{pitch} and θ_{yaw} can be calculated as follows:

$$\theta_{pitch} = \frac{\Delta x_1}{2f} \quad \text{and} \quad \theta_{yaw} = \frac{\Delta y_1}{2f} \quad (9)$$

As PZT actuators are driven by the drivers to the opposite directions of the detected drifted angles, the angle of the right-angle prism mirror will be changed so as to remain the focused spot of laser beam at the central of QPD1. The mathematical analysis of the beam vector for beam drift compensation can be found in [2] and is not repeated in this paper for the sake of brevity.

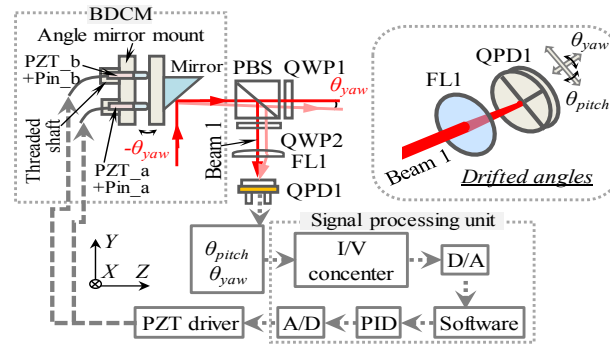


Fig. 4. Compensation principle for laser beam drift.

3. Testing of roll angle measurement system

A prototype of the RAMS was constructed. Its measurement accuracy was tested and the effectiveness of error compensation was investigated. All experiments were conducted in a non-environmental controlled open-type laboratory.

3.1 Stability of measurement system

A compact and low-cost laser diode (DA635, Huanic, China) with a divergence angle less than 0.2 mrad was adopted as the laser source of the optical sensor head. An autocollimator set, consisting of a high precision QPD (QPD1, QP5.8-6-TO5, First Sensor, Germany) and a focus lens (FL1, $\phi 10$, Hitachi, Japan), was constructed to detect θ_{pitch} and θ_{yaw} . A commercial autocollimator (5000U3050, AutoMat, China), having a measurement accuracy of 0.2 arcsec and a repeatability of 0.05 arcsec, was used to calibrate the performance of the autocollimator set. Figure 5 shows the calibrated results. It can be seen that the residuals of θ_{pitch} and θ_{yaw} were all within ± 0.5 arcsec in the measurement range of ± 100 arcsec.

A BDCM, which was primarily composed of an angle mirror mount (GCM080205M, DHT Co., China) and two PZT actuators ($AL1.65 \times 1.65 \times 5D-4F$, NEC TOKIN Electronic, Japan), was instrumented to improve the stability of the measurement system. Figure 6 shows the stability of the roll angular error with and without the compensation of beam drift over a duration of 30 min. The sampling frequency was set to be 100 Hz and a Butterworth low-pass filter was used to reduce the noise level. The distance between the optical sensor head and the detecting unit was set to be 1 m. As seen, the stability was improved from 3.0 arcsec before compensation to 0.6 arcsec after compensation. The effectiveness of the BDCM was verified.

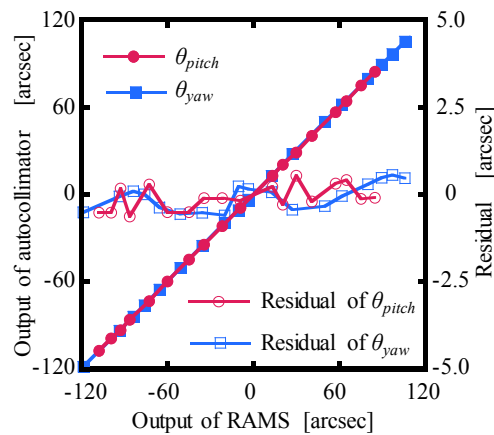


Fig. 5. Calibration results of QPD1 for measurements of drifted angles.

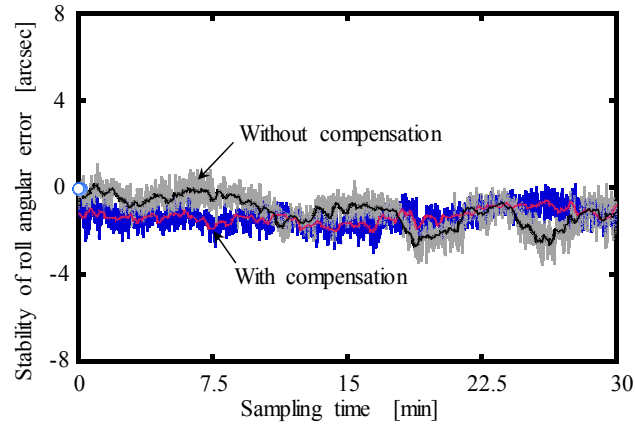


Fig. 6. Stability of the designed system with and without beam drift compensation.

3.2 Sensitivity correction of detectors

Two high precision QPDs (QP50-6-TO8, First Sensor, Germany), for QPD2 and QPD3, with an active area of 11.78 mm per element, an element gap of 42 μm and responsivity of 0.40 A/W, were chosen as the straightness and roll angular errors' detectors. The sensitivity of QPD2 and QPD3 at d_o and d_f were firstly calibrated by a commercial digital indicator (P12D HR, Sylvac, Switzerland) having a resolution of 0.01 μm and a measurement accuracy of 0.22 μm . During the installation of the RAMS, d_o and d_f were set to be 80 mm and 1160 mm, respectively.

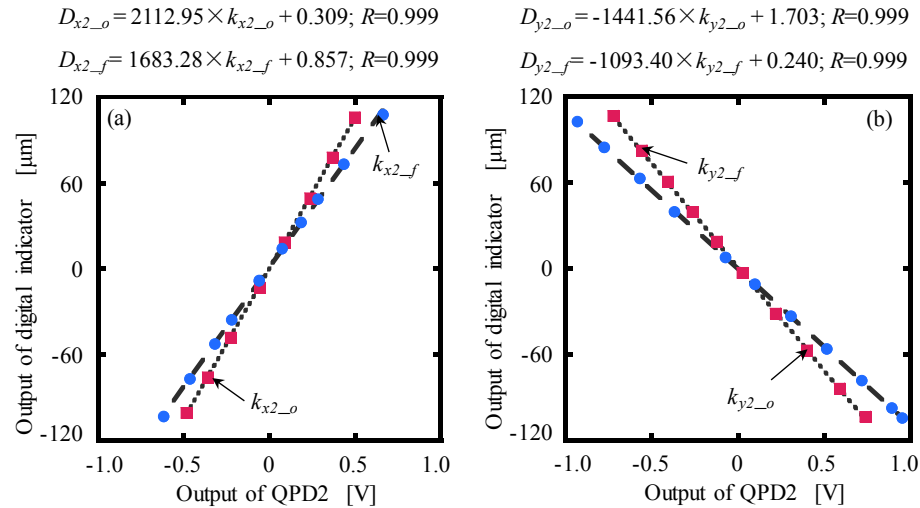


Fig. 7. Calibration results of QPD2 at d_o and d_f in the (a) X -direction and (b) Y -direction.

Figures 7(a) and 7(b) show the relationship between the output voltages of QPD2 and the output displacements of the digital indicator at d_o and d_f in the X - and Y -directions, respectively. The slope of each curve, which was fitted by the least squares method, represents the sensitivity of QPD2. The X -directional sensitivity at d_o and d_f were evaluated to be 2112.95 $\mu\text{m}/\text{V}$ and 1683.28 $\mu\text{m}/\text{V}$, respectively. k_{y2_o} and k_{y2_f} representing the Y -directional sensitivity at d_o and d_f , were evaluated to be -1441.56 $\mu\text{m}/\text{V}$ and -1093.40 $\mu\text{m}/\text{V}$, respectively. The linear correlation coefficient of each sensitivity was calculated to be 0.999.

As shown in the figures that the sensitivity did vary with the change of the measurement distance.

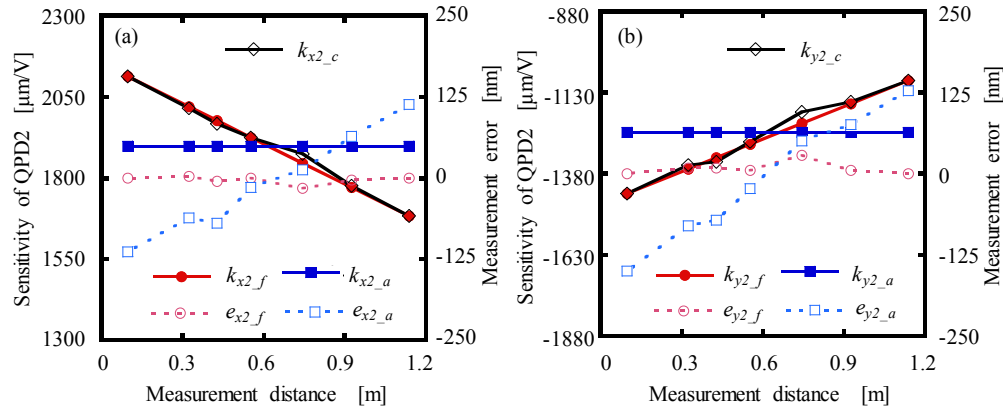


Fig. 8. Sensitivity of QPD2 and measurement errors caused by sensitivity variation in the (a) X-direction and (b) Y-direction.

Then, the QPD2's sensitivity k_{m2_f} ($m = x, y$) at any other position between d_o and d_f was corrected by substituting the calibrated results at d_o and d_f into Eq. (4), as shown in Fig. 8. For comparison, k_{m2_c} and k_{m2_a} were also plotted in the figures. k_{m2_c} represents the actual sensitivity of QPD2 at a specified position. k_{m2_a} , which was conventionally used in the QPD-based measurements [2,10–15], was an average value of k_{m2_c} at different positions over the entire measurement range. It can be seen that k_{m2_f} is in a good agreement with k_{m2_c} . The errors caused by inaccuracy of QPD sensitivity at each target position to one micrometer long are also shown in Fig. 8. The error caused by inaccuracy of k_{x2_f} , referred to as e_{x2_f} , was evaluated to between 2 nm to −15 nm and the error e_{x2_a} corresponding to k_{x2_a} was approximately ± 112 nm. Figure 8(b) shows the results of e_{y2_f} and e_{y2_a} , which were evaluated to between 0 nm and 28 nm, and between 127 nm and −150 nm, respectively.

The sensitivities of QPD 3 in the X- and Y-directions and the corresponding measurement errors are shown in Fig. 9. e_{x3_f} was between −0.7 nm and 35 nm, and e_{y3_f} was between 11 nm and −14 nm. Such errors (e_{mi_f} , $m = x, y$ and $i = 2, 3$), shown in Fig. 8 and Fig. 9, are small enough compared to 1 μm and is deemed negligible, from which the effectiveness of the proposed compensation method for the error caused by detector sensitivity variation can be demonstrated.

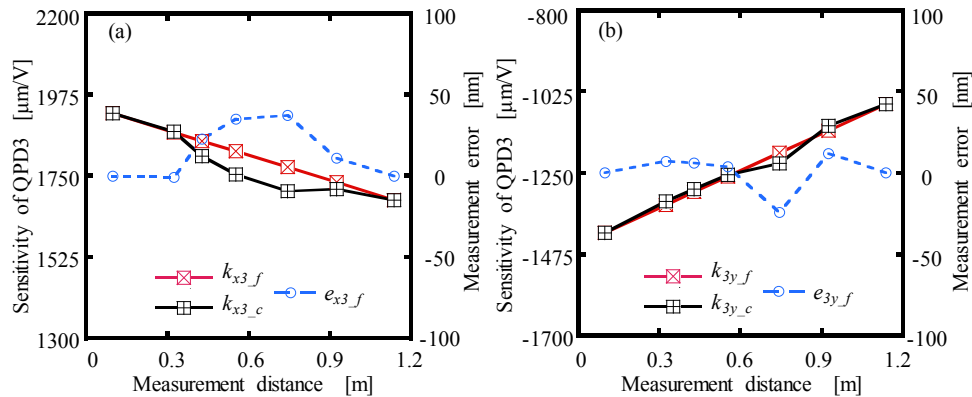


Fig. 9. Sensitivity of QPD3 and measurement errors caused by sensitivity variation in the (a) X-direction and (b) Y-direction.

3.3 Roll angular error measurements of linear stage

The performance investigations of the designed RAMS for measuring the roll angular error of a linear stage were conducted. The travel of the moving table was 1160 mm. The distance L between beam 2 and beam 3 was set to be 120 mm. The resolution of QPD2 and QPD3 were verified to be $0.1 \mu\text{m}$. The theoretical resolution for the roll angle measurement was, thus, evaluated to be 0.17 arcsec .

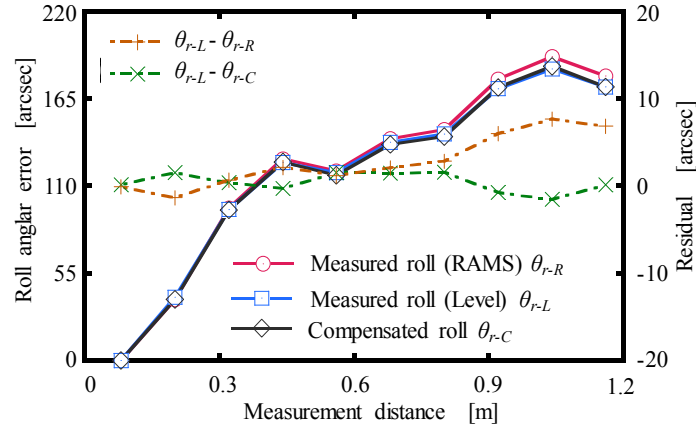


Fig. 10. Measured results of roll angular error with and without compensation.

The roll angular error measurement experiments of the linear stage were performed by the comparison between the developed RAMS and a high precision commercial electronic level (WL/AL11, Qianshao, China), which has a measurement accuracy of 0.2 arcsec . The detecting unit and the electronic level were mounted on the moving table of the linear stage. Figure 10 shows the measurement results. It should be noted that the sensitivity of QPD2 and QPD3 used in Fig. 10 were corrected by Eq. (4). Due to the existence of two beams non-parallel error θ_p , there was a residual of 7.63 arcsec between the outputs of the RAMS (θ_{r-R}) and the electronic level (θ_{r-L}). The output of the RAMS was then compensated by Eq. (8) and the compensated results were also plotted in Fig. 10. It can be seen that the compensated results of the RAMS (θ_{r-C}) were in a good agreement with the outputs of the electronic level. The maximum residual was evaluated to be within $\pm 1.1 \text{ arcsec}$. It has also been confirmed that the roll angular error measurement has a good repeatability and the maximum residuals for six measurements were calculated to be within $\pm 1.2 \text{ arcsec}$ in the measurement distance up to 1 m although the figures are omitted for the sake of clarity. Therefore, the designed system has a satisfied accuracy for roll angular error measurement of precision machines.

4. Conclusions

The design of a roll angle measurement system (RAMS), which is composed of an optical sensor head and a detecting unit, has been presented. The RAMS can be employed to measure both the straightness errors and the roll angular error of precision machines, being a 3DOF error measurement system. A low-cost laser diode was adopted as the laser source. A beam drift compensation module (BDCM) has been constructed and employed to improve the stability of the measurement system. The correction scheme for detector sensitivity variation along the travel has been applied. The measurement error caused by non-parallelism of dual-beam has been compensated. These three special features highlight the performance of the RAMS and make the robust measurement possible. The feasibility of the designed measurement system and the effectiveness of the proposed compensation methods have been verified by a series of calibration and comparison experiments. It has been confirmed that the roll angle measurement accuracy of the designed RAMS was $\pm 1.2 \text{ arcsec}$ in the measurement

distance up to 1 m. In the future work, the measurement uncertainty analysis of the system will be conducted. Moreover, the expansion of the designed system to a 5DOF measurement system (5DOFMS) or a 6DOFMS for measuring the five/six geometric errors of precision machines will be implemented in the future work.

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